

Enhancing Sign Language Recognition Using CNN and SIFT: A Case Study on Pakistan Sign Language

Overview

The study addresses the communication barrier faced by the deaf and hard-of-hearing community by proposing an efficient computer vision-based system to recognize Pakistan Sign Language (PSL). It leverages Scale-Invariant Feature Transform (SIFT) for feature extraction and a Convolutional Neural Network (CNN) for classification of Urdu alphabet-based static hand gestures. This hybrid method aims to enhance recognition accuracy under varying lighting conditions and with minimal error.

Methodology

Data Collection

Dataset Size: 2220 images

Subjects: 20 participants (10 male, 10 female), aged 20–40

Environment: Captured using Kinect motion sensor under three lighting conditions (bright, moderate, dim) and at varying distances (2ft, 3ft, 4ft)

Focus: 37 Urdu alphabet gestures (excluding one dynamic gesture)

Preprocessing

Grayscale Conversion: Using luminosity method

Histogram Equalization: For contrast enhancement

Segmentation: Thresholding to isolate hand region from the background

Feature Extraction

Technique: SIFT

Features Extracted:

Hand margin/edges

Hand size and shape (bounding box area)

Palm center point

Finger positions (distance and angle from palm center)

Descriptor: Converts features to vector points

Classification

Model: CNN with 5 convolution layers, 5 pooling layers, 2 fully connected layers

Framework: Implemented in Python with TensorFlow on Google Colab

Training Parameters:

Batch size: 32

Epochs: 100

Activation: ReLU (hidden layers), Softmax (output)

Data split: 70% training, 30% testing

Results

Training Accuracy: 99.65%

Validation Accuracy: 98.74%

Error Rate: 1.26%

Confusion Matrix: Demonstrates high class-wise accuracy across all 37 alphabet signs

Best Image Resolution for Processing: 1920×1080 under moderate light

Overfitting Observed: When training was extended to 200 epochs, validation accuracy dropped, indicating the importance of early stopping

Comparative

Technique	Accuracy (%)	Error Rate (%)	Dataset Size	Notes
Proposed (SIFT+CNN)	98.74	1.26	2220	Best performance overall
SVM (Shah et al.)	91.98	8.02	6633	Lower accuracy
CNN (Saleem et al.)	95.00	5.00	1502	Pure CNN
DCCNN (Wu, 2020)	98.02	1.98	720	Smaller dataset

Conclusion

The proposed framework, combining SIFT-based feature extraction with CNN classification, significantly enhances the recognition of Pakistan Sign Language gestures. It demonstrates high accuracy, low error, and strong adaptability to varying lighting conditions. This system outperforms many existing methods and provides a robust basis for future development in real-time, video-based PSL recognition and multilingual extensions.